

Table C.1: Heat Capacities of Gases in the Ideal-Gas State[†]

Constants in equation $C_P^{ig}/R = A + BT + CT^2 + DT^{-2}$ for T (K) from 298 K to $T_{\max}$

Chemical species		$T_{\max}$	$C_{P_{298}}^{ig}/R$	A	$10^3 B$	$10^6 C$	$10^{-5} D$
Alkanes:							
Methane	CH ₄	1500	4.217	1.702	9.081	-2.164	
Ethane	C ₂ H ₆	1500	6.369	1.131	19.225	-5.561	
Propane	C ₃ H ₈	1500	9.011	1.213	28.785	-8.824	
<i>n</i> -Butane	C ₄ H ₁₀	1500	11.928	1.935	36.915	-11.402	
<i>iso</i> -Butane	C ₄ H ₁₀	1500	11.901	1.677	37.853	-11.945	
<i>n</i> -Pentane	C ₅ H ₁₂	1500	14.731	2.464	45.351	-14.111	
<i>n</i> -Hexane	C ₆ H ₁₄	1500	17.550	3.025	53.722	-16.791	
<i>n</i> -Heptane	C ₇ H ₁₆	1500	20.361	3.570	62.127	-19.486	
<i>n</i> -Octane	C ₈ H ₁₈	1500	23.174	4.108	70.567	-22.208	
1-Alkenes:							
Ethylene	C ₂ H ₄	1500	5.325	1.424	14.394	-4.392	
Propylene	C ₃ H ₆	1500	7.792	1.637	22.706	-6.915	
1-Butene	C ₄ H ₈	1500	10.520	1.967	31.630	-9.873	
1-Pentene	C ₅ H ₁₀	1500	13.437	2.691	39.753	-12.447	
1-Hexene	C ₆ H ₁₂	1500	16.240	3.220	48.189	-15.157	
1-Heptene	C ₇ H ₁₄	1500	19.053	3.768	56.588	-17.847	
1-Octene	C ₈ H ₁₆	1500	21.868	4.324	64.960	-20.521	
Miscellaneous organics:							
Acetaldehyde	C ₂ H ₄ O	1000	6.506	1.693	17.978	-6.158	
Acetylene	C ₂ H ₂	1500	5.253	6.132	1.952		-1.299
Benzene	C ₆ H ₆	1500	10.259	-0.206	39.064	-13.301	
1,3-Butadiene	C ₄ H ₆	1500	10.720	2.734	26.786	-8.882	
Cyclohexane	C ₆ H ₁₂	1500	13.121	-3.876	63.249	-20.928	
Ethanol	C ₂ H ₆ O	1500	8.948	3.518	20.001	-6.002	
Ethylbenzene	C ₈ H ₁₀	1500	15.993	1.124	55.380	-18.476	
Ethylene oxide	C ₂ H ₄ O	1000	5.784	-0.385	23.463	-9.296	
Formaldehyde	CH ₂ O	1500	4.191	2.264	7.022	-1.877	
Methanol	CH ₄ O	1500	5.547	2.211	12.216	-3.450	
Styrene	C ₈ H ₈	1500	15.534	2.050	50.192	-16.662	
Toluene	C ₇ H ₈	1500	12.922	0.290	47.052	-15.716	
Miscellaneous inorganics:							
Air		2000	3.509	3.355	0.575		-0.016
Ammonia	NH ₃	1800	4.269	3.578	3.020		-0.186
Bromine	Br ₂	3000	4.337	4.493	0.056		-0.154
Carbon monoxide	CO	2500	3.507	3.376	0.557		-0.031
Carbon dioxide	CO ₂	2000	4.467	5.457	1.045		-1.157
Carbon disulfide	CS ₂	1800	5.532	6.311	0.805		-0.906
Chlorine	Cl ₂	3000	4.082	4.442	0.089		-0.344
Hydrogen	H ₂	3000	3.468	3.249	0.422		0.083
Hydrogen sulfide	H ₂ S	2300	4.114	3.931	1.490		-0.232
Hydrogen chloride	HCl	2000	3.512	3.156	0.623		0.151
Hydrogen cyanide	HCN	2500	4.326	4.736	1.359		-0.725
Nitrogen	N ₂	2000	3.502	3.280	0.593		0.040
Nitrous oxide	N ₂ O	2000	4.646	5.328	1.214		-0.928
Nitric oxide	NO	2000	3.590	3.387	0.629		0.014
Nitrogen dioxide	NO ₂	2000	4.447	4.982	1.195		-0.792
Dinitrogen tetroxide	N ₂ O ₄	2000	9.198	11.660	2.257		-2.787
Oxygen	O ₂	2000	3.535	3.639	0.506		-0.227
Sulfur dioxide	SO ₂	2000	4.796	5.699	0.801		-1.015
Sulfur trioxide	SO ₃	2000	6.094	8.060	1.056		-2.028
Water	H ₂ O	2000	4.038	3.470	1.450		0.121

[†]Selected from H. M. Spencer, *Ind. Eng. Chem.*, vol. 40, pp. 2152–2154, 1948; K. K. Kelley, *U.S. Bur. Mines Bull.* 584, 1960; L. B. Pankratz, *U.S. Bur. Mines Bull.* 672, 1982.

Table C.2: Heat Capacities of Solids[†]

Constants for the equation $C_p/R = A + BT + DT^{-2}$ for T (K) from 298 K to $T_{\max}$

Chemical species	$T_{\max}$	$C_{P_{298}}^{ig}/R$	A	$10^3 B$	$10^{-5} D$
CaO	2000	5.058	6.104	0.443	-1.047
CaCO ₃	1200	9.848	12.572	2.637	-3.120
Ca(OH) ₂	700	11.217	9.597	5.435	
CaC ₂	720	7.508	8.254	1.429	-1.042
CaCl ₂	1055	8.762	8.646	1.530	-0.302
C (graphite)	2000	1.026	1.771	0.771	-0.867
Cu	1357	2.959	2.677	0.815	0.035
CuO	1400	5.087	5.780	0.973	-0.874
Fe(α)	1043	3.005	-0.111	6.111	1.150
Fe ₂ O ₃	960	12.480	11.812	9.697	-1.976
Fe ₃ O ₄	850	18.138	9.594	27.112	0.409
FeS	411	6.573	2.612	13.286	
I ₂	386.8	6.929	6.481	1.502	
LiCl	800	5.778	5.257	2.476	-0.193
NH ₄ Cl	458	10.741	5.939	16.105	
Na	371	3.386	1.988	4.688	
NaCl	1073	6.111	5.526	1.963	
NaOH	566	7.177	0.121	16.316	1.948
NaHCO ₃	400	10.539	5.128	18.148	
S (rhombic)	368.3	3.748	4.114	-1.728	-0.783
SiO ₂ (quartz)	847	5.345	4.871	5.365	-1.001

[†]Selected from K. K. Kelley, *U.S. Bur. Mines Bull.* 584, 1960; L. B. Pankratz, *U.S. Bur. Mines Bull.* 672, 1982.

Table C.3: Heat Capacities of Liquids[†]

Constants for the equation $C_p/R = A + BT + CT^2$ for T from 273.15 to 373.15 K

Chemical species	$C_{P_{298}}^{ig}/R$	A	$10^3 B$	$10^6 C$
Ammonia	9.718	22.626	-100.75	192.71
Aniline	23.070	15.819	29.03	-15.80
Benzene	16.157	-0.747	67.96	-37.78
1,3-Butadiene	14.779	22.711	-87.96	205.79
Carbon tetrachloride	15.751	21.155	-48.28	101.14
Chlorobenzene	18.240	11.278	32.86	-31.90
Chloroform	13.806	19.215	-42.89	83.01
Cyclohexane	18.737	-9.048	141.38	-161.62
Ethanol	13.444	33.866	-172.60	349.17
Ethylene oxide	10.590	21.039	-86.41	172.28
Methanol	9.798	13.431	-51.28	131.13
<i>n</i> -Propanol	16.921	41.653	-210.32	427.20
Sulfur trioxide	30.408	-2.930	137.08	-84.73
Toluene	18.611	15.133	6.79	16.35
Water	9.069	8.712	1.25	-0.18

[†]Based on correlations presented by J. W. Miller Jr., G. R. Schorr, and C. L. Yaws, *Chem. Eng.*, vol. 83(23), p. 129, 1976.